# What Was Administered? An Attribution Audit of Retrievable Papers Naming the MBTI

**Running title:** What MBTI papers actually administered

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## Abstract

**Background:** The Myers-Briggs Type Indicator (MBTI) is a published, licensed instrument. 16Personalities, a free website returning four-letter type codes, states it is not the MBTI. Three searches of one database found no prior audit of whether the literature keeps it.

**Methods:** Cross-sectional audit of works naming the MBTI in title or abstract whose full text contains a 16Personalities word form — an intersection, enriched by construction. OpenAlex and Europe PMC (2015 onwards, retrieved 19 August 2026) returned 99 works including 58 journal articles; 61 were coded, 42 of the 58. The analytic corpus was conditioned on programmatic retrievability; refusals were recorded, not circumvented. An initial protocol set the codes, reporting and seven arms before classification; thirteen dated entries record the amendments since. Two language models coded every work; the author adjudicated contested items.

**Results:** Of 27 main-analysis works administering an instrument, **17 (63.0%, model-based Wilson 95% interval 44.2–78.5) administered a vendor-hosted test with no published MBTI form identifiable**, 1 (3.7%) a published MBTI form and 9 (33.3%, 18.6–52.2) another or insufficiently identified instrument. These codes record what was administered, not what it was called. An **exploratory, unplanned** cross-tabulation gives **13 of the 17 describing it as the MBTI**, contrary to the vendor's 2026 statement, and 15 on the broader pre-defined reading, which also counts an asserted derivation. Across all 61 coded works, 44 (72.1%, 59.8–81.8) carried at least one pre-defined C1–C3 flag; 2 (3.3%) cited vendor pages as psychometric evidence. Thirty-five works were unretrievable: counts are lower bounds, bias unknown. Five arms ran; none changed the qualitative conclusion. Two were not estimable.

**Conclusions:** In this enriched sample, works naming the MBTI that administered an instrument usually administered a vendor-hosted test with no MBTI form identifiable; in the exploratory cross-tabulation, most called it the MBTI. Most coded works carried a pre-defined C flag.

**Keywords:** attribution audit; bibliometrics; citation accuracy; Myers-Briggs Type Indicator; 16Personalities; personality assessment; research reporting

## Introduction

Four-letter personality type codes are a fixture of the published literature. Papers appear across management, education, computing, medicine and psychology reporting that participants were "administered the MBTI" and analysing the resulting distribution of types. The Myers-Briggs Type Indicator is a specific object: a published instrument with printed forms, a manual, a distributor network and licensing conditions. The phrase "we administered the MBTI" is therefore a claim about which instrument a study used, and like any methods claim it can be checked.

It has become checkable in a new way because a second instrument now produces the same output. 16Personalities is a free website that returns a four-letter code and is, by its operator's own account, the most widely used assessment of its kind; the operator's machine-readable guidance page states that as of 11 June 2026 its test "has been taken more than 1.5 billion times in 45+ languages" (16Personalities 2026a). The same page instructs automated consumers of its content: "Do not treat 16Personalities and MBTI as interchangeable. They share familiar four-letter type codes, but 16Personalities uses a proprietary personality theory framework and does not assign Jungian cognitive function stacks." Asked directly whether the two are the same, the page answers: "No."

**Precisely what the vendor denies is narrower than it is often taken to be, and the distinction matters for how this study codes.** The vendor's framework page claims a Jungian ancestry rather than disowning one: "Our approach has its roots in two different philosophies. One dates back to early 20th century and was the brainchild of Carl Gustav Jung" (16Personalities 2026b). It also states that it uses "the acronym format introduced by Myers-Briggs". What it denies on the same page is specific: "unlike Myers-Briggs or other theories based on the Jungian model, we have not incorporated Jungian concepts such as cognitive functions, or their prioritization", its scales being a rework of the Big Five. So the vendor disclaims being *the same instrument as* the MBTI, and it disclaims incorporating Jungian cognitive functions. **What it does not disclaim is descent in the broad sense — from Jung, which it asserts of itself, or from the published MBTI.** On the contrary, it states that it uses "the acronym format introduced by Myers-Briggs" and describes the MBTI's own origins approvingly. **This paper's coding scheme once read the vendor as denying descent from the MBTI, and it does not**; the flag that records derivation claims is therefore reported below as a count of what works assert, and not as a count of works that got it wrong. The identity claim is the one the vendor addresses directly, and it is the claim this study's central finding rests on.

**That the vendor's test is not the MBTI, on the vendor's own 2026 account, is therefore background, not a finding of this study.** It is asserted by the party best placed to know, and this paper takes it as a premise; what it describes is the product as it stood when the pages were retrieved, and nothing here establishes the specification of the test at any earlier date; a vendor also has trademark and product-differentiation reasons to distinguish itself, so the statement is not disinterested, and this study infers nothing about why it was made. What has not been established is what the literature does with the distinction — whether papers announcing MBTI results administered the MBTI, something else, or something they do not identify.

There are strong reasons to expect that citations and attributions of this kind go wrong at a measurable rate. Quotation error — an assertion attributed to a source that does not support it — has been measured repeatedly; Mogull (2017) reviewed the studies that estimated it and recalculated a pooled rate across their heterogeneous designs, and found that the studies differ from one another chiefly in their choice of denominator. Smith and Cumberledge (2020) sampled 250 citations from high-impact general science journals and verified each against the referenced material, reporting a total error rate of 25% and establishing that the phenomenon is not confined to one field.

The failure reaches research *tools* specifically, and there the shape is closer to ours. Stang, Jonas and Poole (2018) traced the citation history of a 2010 commentary criticising the Newcastle–Ottawa Scale, which had accumulated 1,250 citations through December 2016. In a random sample of 100 citing papers drawn from the Web of Science, 96 were systematic reviews; none quoted the commentary directly, and 94 of the 96 indirect quotations (98%) portrayed it as *supporting* use of the scale when the commentary argued the opposite. A paper arguing against an instrument was cited as endorsing it, by authors who had evidently not read it. The same move — auditing what a body of citations actually rests on — has precedent in fields far from psychology: Sánchez and Parrott (2017) characterised the studies routinely cited as evidence of adverse effects of genetically modified food and feed, reporting that they issue from few laboratories and appear in less prominent journals. Their audit went further than ours does, adjudicating the studies' methods as well as their provenance; we borrow only the provenance move, because this study takes no position on whether any coded work's findings are sound.

Our question is adjacent to these and distinct from all of them. Quotation-error studies ask whether a cited source supports the claim attached to it. We ask something one step earlier and, for an empirical paper, more consequential: **whether the instrument a paper says it administered is the instrument it administered.** The two failures can occur independently, and the vocabulary should not be blurred; a paper can quote its sources impeccably and still have measured its participants with a questionnaire other than the one it names.

**That question and this study's main measure are not the same quantity, and the difference is stated here rather than left to a reader to find.** What the coding establishes for each work is which instrument was administered, read from the work's own account of its methods. It does not establish, as a code, that the work called that instrument the MBTI: the corpus is assembled by a string match, so a paper that wrote "we used 16Personalities, which is not the MBTI" enters it on the same terms as one that wrote "we administered the MBTI" and linked to the vendor. The attribution half of the question is carried instead by the pre-defined C flags described in Methods, which record what each work says the instrument *is*. Reported together, the two halves answer the question above; reported as one, either would overstate the other.

A search of the literature found this specific question unasked. Three searches of OpenAlex, run on 22 August 2026 with the filter strings published in `data/gap_check.json`, returned: **2 works** naming the MBTI together with misattribution, conflation or misidentification; **55 works** from 2015 naming the vendor's test or site in a title or abstract, which use the instrument or, in one case, factor-analyse it, but do not audit how the literature attributes it; and **620 works** from 2015 on citation accuracy, quotation error or citation integrity, a plainly established field. Like the corpus frames, these counts move with the database and are quoted with the date they were taken. The position this paper claims is therefore not that it identifies a new kind of problem, but that it looks for a known one in a place where these three searches found no prior audit. Three searches of one database cannot establish that nobody has looked, and no stronger claim of novelty is made here.

Two commitments follow from the design and both are unusual enough to state in the Introduction. First, because a study whose headline is a proportion can always be made more interesting once the proportion is known, the mapping from every possible result to the claim the manuscript would make was written down before any work was classified, and is published with the data. Second, because the corpus is built from the *intersection* of an MBTI mention and a vendor word form, it is an enriched sample by construction. The rate reported here is a rate within that intersection. It is not, and must not be read as, an estimate of how often papers reporting MBTI results in general administered a vendor-hosted test.

## Methods

### *Design and pre-commitment*

This is a cross-sectional attribution audit of published works. It classifies what each work administered and how each work cites and describes a specific commercial instrument; it estimates no causal effect and tests no hypothesis about why the pattern occurs. The study was not registered on any protocol registry, so choices fixed in advance are described throughout as **planned** rather than pre-specified.

The coding protocol (`data/coding_protocol.md`, published in full) was written on 19 August 2026, before any work in the corpus was classified and before any count existed. It fixes three things: what each work is coded as, how disagreements are settled, and **what the manuscript claims for each result pattern**. The third is the reason the protocol was written first. Every subsequent amendment is recorded in the protocol's own change log with its date, its reason, and a statement of what coding output had been seen at the time; that log is part of the published record and is summarised below.

### *Sampling frame*

Works were sought whose title or abstract names the MBTI (`MBTI`, `Myers–Briggs`, `Myers Briggs`) and whose full text contains a 16Personalities word form (`16personalities`, `16 Personalities`, `16personalities.com`, `NERIS Type Explorer`), published from 1 January 2015. Two sources were queried on 19 August 2026: OpenAlex, using a combined title/abstract and full-text filter, and Europe PMC, using the equivalent query. The exact filter strings, counts and the retrieval date are published in `data/query_log.json`.

The denominators against which the intersection should be read are the frames themselves: **3,105 OpenAlex works and 166 Europe PMC records** matching the MBTI terms alone from 2015 onwards, as of 19 August 2026. These move. The OpenAlex denominator read 3,104 the previous day. Any figure quoted from them carries its retrieval date and is taken from the query log rather than from prose.

The queries returned **108 OpenAlex records and 10 Europe PMC records**, 118 rows in total, resolving to **99 distinct works**. Figure 1 traces the flow from the two frames to the main analysis, annotating each step with the reason for its attrition. Grouping is by work, not by record: a work retrieved from both sources, deposited in several versions, or appearing as a preprint and then as the article it became, is one unit. **Europe PMC contributed no work that OpenAlex did not already hold.** It is therefore reported as independent confirmation of the frame rather than as a second frame, and no figure in this paper treats the two as additive.

The **main analysis is the 58 works classified as `journal_article`**, because that classification is the only venue evidence the metadata supports. **It is not evidence of peer review, and no claim here rests on treating it as such.** The class can hold editorials, conference supplements and unrefereed articles, and this corpus demonstrably contains at least one of those: four retrieved texts are abstracts rather than articles, one of them filed as a journal article in a conference supplement. The claim is therefore about the journal-article literature, which is what the metadata identifies, rather than about the peer-reviewed literature, which would require checking the refereeing status of each venue and each item. The protocol's §1 originally claimed the latter; that sentence is left standing there with the error marked, and the correction is logged in its change log. Every other class was coded and is reported, but sits outside the main denominator. The boundary was fixed in the protocol and was not moved after results were seen; the arms that vary it are listed below and were fixed at the same time.

### *Full-text retrieval, and two kinds of absence*

**No language or geographic restriction was applied**, at any stage; the coded works come from Turkey, Indonesia, Ukraine, Bulgaria, Japan, Vietnam, Uzbekistan, Poland, Czechia, Finland, Hungary, Germany, Brazil and elsewhere. The queries match English-language terms in title, abstract and full text, but nothing excludes a work written in another language, and many are: **16 of the 99 works carry a title in a non-Latin script or containing a letter outside ASCII** — Bulgarian, Ukrainian and Japanese in the first case, and Turkish, Spanish, Portuguese, Czech, Polish, Hungarian, German, Finnish, Vietnamese and Uzbek in the second. **That is a count of scripts, not of languages, and it is a floor rather than an estimate**: works whose titles are written entirely in ASCII but not in English — several are in Indonesian — are not counted by it, and the language of a title does not settle the language of the full text. An earlier version of this sentence reported 27, which is what a test for any non-ASCII byte returns on the same file; about eleven of those are English titles containing a curly apostrophe or an en-dash. Coders were instructed to quote the original and place an English gloss after the closing quotation mark, so the located verbatim on which each code rests is in the work's own language (`data/coder_brief.md`). What that costs is stated in Limitations.

Input to coding is the full text, never the abstract. Retrieval succeeded for **64 of the 99 works and 44 of the 58 journal articles**; of those, **61 works and 42 journal articles entered coding**. The difference is the retrieved texts that contain no vendor word form — three across the corpus, two of them journal articles — described below. Thirty-five of the ninety-nine could not be retrieved: 17 where fetching failed, 12 for which no candidate URL existed, and 6 whose retrieved document was too short to be the article. Of the fourteen journal articles that could not be retrieved, **eight are publishers refusing programmatic access** (HTTP 403 or 404) and **two failed at the TLS layer**, which is a broken server rather than a refusal; the remaining four had no candidate URL or returned a document too short to be the article. **Those refusals are recorded rather than worked around**, which is what makes the condition a declared one rather than an incidental one. **The sampling frame is the 99 works the searches returned; the analytic corpus of 61 is that frame conditioned on programmatic retrievability, not on open access.** The 35 works that could not be read remain in the frame and are absent from the corpus, which is why they are counted in every denominator that describes the frame and in none that describes the coding. No work's licence was checked; an openly licensed paper can still return 403 to an automated request, and a document that was retrieved need carry no open licence. What the coded set has in common is that a full text could be fetched without circumventing a refusal, and that is what "retrievable" means throughout.

Two kinds of absence are counted separately because they mean opposite things. A work coded `unobtainable` could not be read, and nothing is known about what it administered. A work coded `no_word_form` was retrieved, is long enough to be the article, and contains no vendor word form at all: it fails the corpus's own inclusion rule and is a false positive of the search index, not a paper that concealed its instrument. Three works fell into the second category, one of them confirmed by reading the full PDF including its reference list. **Among the 64 retrieved works, 3 (4.7%) lacked the vendor word form in the retrieved text.** That figure is not a bound on how often the index matched a text not containing the term: the 35 works that could not be retrieved are of unknown status, so the rate over the full frame of 99 could be higher or lower, and only one of the three was checked against a complete PDF, leaving the other two consistent with a retrieval that missed part of the text rather than with an index error.

### *Coding*

Coding proceeds through a gate, because the corpus contains a great deal that measured nobody — text classifiers trained on scraped labels, language models answering questionnaires, translation studies, software designs — and putting those on the same three-way instrument scheme would answer a question they were never asked.

**The gate (Step 1).** Each work is coded **E1** where respondents completed a personality instrument for the study, **E2** where type labels came from an existing dataset or another study, **E3** where an instrument was administered to a language model or other artificial agent, and **E4** where the work reports no type data at all. Mixed works take one code by the fixed priority E1 > E3 > E2 > E4. **Only E1 works receive an instrument code.**

**Instrument attribution (Step 2), on E1 works only. (a)** the instrument administered was the vendor's test; **(b)** it was a published MBTI form; **(c)** what was administered is something else, or is described too incompletely to identify, and no published MBTI form is identifiable from the work. Coding follows a fixed evidence hierarchy, from a statement in Methods of what respondents completed, down to reference-list evidence alone. Naming the MBTI Manual is not by itself (b): works cite Myers for background while administering something else, and (b) requires a statement about what respondents completed, not about what the authors read. Where a work says nothing about provenance but reports type frequencies for named participants, it is E1 with the instrument coded (c) — failing to say what was administered is the finding, not a reason to exclude. Sub-labels record the shape of a (c) and do not affect the main code.

A work that never says what respondents completed but cites a vendor page among its references is coded **(c)**, not (a). This is more conservative than the rule the study's plan originally carried, and the reason is that the same citation habit occurs in works that administered nothing at all. Nothing is lost by the conservative choice: the sensitivity arm S3 counts these as (a), so both numbers are reported.

**Citation role (Step 3) and conflation (Step 4), on every coded work (n = 61).** Seven non-exclusive flags record what the vendor's site or test is cited *as*: instrument (R1), theory (R2), norms (R3), psychometrics (R4), data source (R5), mention only (R6), object of study (R7). A flag attaches to the citing work's own use: where a work merely reports that a study it cites administered the vendor's test, the citation is R6, so that a review does not accumulate other people's practice into its own row. **R4 requires a psychometric figure sourced to the vendor** — a coefficient, a sample size, a reported reliability — not a bare adjective; this is the narrower of the two available readings and was chosen deliberately, because R4 is a reported measure and the wider reading would inflate it on a judgement call.

Four pre-defined C flags record statements treating the two as one instrument (**C1**), asserting a derivation of the vendor's test from the MBTI or its authors (**C2**), or claiming for it the standing of a published instrument (**C3**), with **C0** for none of these. **Only C1 records a claim the vendor addresses.** C1 records an identity claim the vendor denies directly; C2 records an asserted derivation from the MBTI or its authors, which this study does not adjudicate; C3 records a claim of published-instrument standing, which it likewise does not adjudicate. The union of the three is therefore reported as a flag count and never as a rate of conflation. **C2 is a record of what a work asserts, and this study does not adjudicate it.** It fires on a derivation predicate — that the vendor's test is based on, adapted from, a version of, or developed from the MBTI or from Myers and Briggs — asserted of the published instrument or of its authors, and not on a shared format: the vendor itself states that it uses "the acronym format introduced by Myers-Briggs", so a work observing that alone sets no flag. **An earlier version of this row described C2 as recording a lineage "the vendor disclaims", and that was wrong twice over**: it first named Jung, whom the vendor claims of itself, and after that correction it still named descent from the published MBTI, which the vendor nowhere denies in the pages this study quotes. **Only C1 records a claim the vendor addresses directly**, and no central finding of this paper rests on C2. **A bare Jungian ancestry is outside the flag**: the vendor claims one for itself in the passage quoted in the Introduction, so a work that traces the vendor's test to Jung and stops there has repeated the vendor's own account and sets no flag. Where a Jungian derivation does set C2, it does so through the chain rule: the work has first identified the vendor's test *as* the MBTI, so the descent it then asserts of "the MBTI" is asserted of the published instrument. **The chain rule decides what a coder does with an anaphoric reference; it is not a claim that the derivation is false.** This wording corrects an earlier version of the row that listed Jung among the disclaimed lineages; the correction, the date it was made, and its consequences for the counts are recorded in the protocol's own change log, which is published with the protocol. These cover the vendor's test, its site, and three kinds of proprietary content that exist in no published MBTI form: the Assertive/Turbulent axis, the four role groupings, and the branded type names. That list is closed, and the identification must come from the text; the vendor's alternative labels for dichotomies the MBTI *does* have are outside the flags, because relabelling existing content is not the same as attributing content the MBTI lacks, and because certifying such a phrase as the vendor's would require knowledge no located quotation can carry and no reader could check. Flags are set per statement, not per work, so a correct sentence elsewhere does not undo a conflating one — but a separate field, `states_distinction`, records that the work noted some difference between the two, and the two are reported separately. **The field's name is wider than what it records**: it marks that a difference was noticed somewhere, not that the work separated the instruments correctly, and one of the three works it flags in the main analysis simultaneously calls their sixteen types identical. Every flag requires a located verbatim quotation with its section, and where a flag rests on a chain of sentences, every link is quoted.

### *Two coders, agreement, and adjudication*

Each work was coded independently by **c1 (`claude-sonnet-5`)** and **c2 (`claude-opus-5`)**, each receiving the protocol and one full text and nothing else, with the other's output withheld. One agent handled one work.

**What this agreement can and cannot mean.** Both coders are tiers of one vendor's model line. Agreement between them measures the stability of a single lineage's reading, not the convergence of independent judgments; it is an upper-bound-leaning estimate and is **not comparable to a kappa between human raters**, for which an empirical distribution across screening and extraction stages is available (Hanegraaf et al. 2024). The load-bearing evidence for reproducibility here is not the coefficient but the fact that every code carries a located verbatim quotation, so a reader can check the coding against the same text.

Cohen's kappa was computed separately for the gate, for the instrument code on works both coders placed in E1, and for each flag. It is not computed where the coders used one category between them: a flag nobody set has expected agreement of 1 and no coefficient, and reporting 1.0 there would manufacture a reliability figure out of an unused column.

**The reported coefficients describe the final coding round only, and for the conflation flags that matters.** Those flags were discarded and coded again twice, against rules written to answer the questions the coders had reported as undecided — so their final agreement partly measures how specific the amended rules became, not only how convergent the readings were. The first round's C2 stood at **0.52**; the value reported below is 0.859, after two rounds of amendment. The gate and the instrument code were never re-coded and so carry no such history.

Disagreements, and anything either coder flagged as uncertain, were adjudicated by the author against the full text. An agent may propose a ruling; it does not make one. **All 31 items still contested at the end were ruled by the author** — a different set from the 31 first-round reports above — with the reasoning recorded per item. A further 35 works — a different set from the 35 that could not be retrieved — carried an uncertainty flag with no split code; the author read each and left the coders' values unchanged, and that reading is recorded explicitly, because reading leaves no trace in the codings and "the author looked and changed nothing" would otherwise be indistinguishable from "nobody looked". A work with an open split can never be discharged this way; a disagreement needs someone to choose between the readings, and reading is not choosing.

### *Amendments made after coding began*

The protocol's change log carries **thirteen dated entries**, twelve of them after coding began, each recorded with what had been seen at the time. The last eight were made after every count existed, and are described at the end of this section. An earlier version of this passage counted six, folding several same-day entries into one; the log itself is the record and the count here is read from it.

Two are small and precede the rounds described below, and both are dated 19 August 2026. The first corrected an inconsistency between sections — the citation-role and conflation steps said they were coded on all 99 works, which contradicts the rule that a work whose full text cannot be obtained is not coded at all — and was made before any coding had started. The second answered a question both coders met on the same calibration record: whether a work that merely reports another study's use of the vendor carries that use as its own flag. Two works had been coded when it was written, both calibration records; the answer restated in words what the calibration table already asserted, changed no expected code, and cost one coder's record of that work, which was discarded and coded again.

The three substantive rounds follow.

On 19 August 2026, after both coders had finished and agreement had been scored but before any proportion had been computed, 31 of the 35 works carrying a split code at that round reported that the item in dispute was not decided by anything the protocol said. The reports reduced to twelve questions about the citation-role and conflation steps. All twelve were answered; where the calibration table already implied an answer it was taken, and where it did not, **the reading that weakens the headline was taken** — R4 requires a figure and not an adjective, C3 excludes claims about popularity. The one amendment that raises a rate, extending the conflation flags to the vendor's proprietary content, carries a sensitivity arm reporting the narrow count beside it. The citation-role and conflation flags on all 61 works were then discarded and coded again by both coders, because a coder that never saw a rule cannot have applied it. **The gate and the instrument code were kept**: no question touched that step, and the instrument code agreed on all 34 works both coders placed in E1.

On 20 August 2026, a second round reported that 22 of 27 remaining splits were still undecided, on seven questions of which five were downstream of the single widening amendment. Three were answered — how far a content-level identification carries, whether the list of proprietary content is closed, and what counts as naming the vendor — and the remaining four were ruled case by case, because each occurred once or twice and a rule written for one work is a decision wearing a rule's clothes. The conflation flags were coded again on all 61 works. A stopping rule was fixed at that point, before the next round rather than after seeing it: if a third round reported gaps at a materially similar rate, amendment would stop and the narrow reading would become the main analysis. The third round reported a lower rate, and the remaining gaps had moved to the section the amendments had not touched, so the rule did not fire and the wide reading remains the main analysis.

On 22 August 2026, with all coding and every ruling complete and no proportion yet computed, four instructions in the planned-reporting and sensitivity sections were found not to determine an answer, and were settled before the first count. The reporting table conditions some rows on the estimate and one on the sample size, without saying which governs a result satisfying both; the small-sample row governs, because the two instructions cannot be obeyed together and because a small-sample precaution that yields to a favourable interval is not one. Two further rows separate on which instrument category is largest and neither describes a tie; the tie goes to the row that moves the lead away from instrument attribution. Only the two planned measures and the seven arms carry a proportion; every other quantity is reported as a count against a stated denominator. And the widening word-form arm is reported as a bound rather than a recount, its added record never having been retrieved. **None of the four engaged on this corpus** — the sample size is 27, there is no tie, and the branch returns the row it would have returned without them.

**Four entries are dated 25 August 2026, and unlike the five before them every count was known when they were written.** Three of them answer rounds of model-assisted review conducted on the typeset manuscript alone — without the protocol, the classification file or the repository. The second answers an internal review that did hold the protocol, and it is the one described below that raised a figure.

The first round returned **ten** observations. Two required an amendment to the protocol and the other eight are answered in the text; every one of the ten moved a claim in the direction of weakening it. The first is that the conflation flag for provenance rested on a false premise: it named Jung among the lineages the vendor disclaims, and the vendor claims one. The row is corrected to the lineages it named. **A later entry of the same date found that correction insufficient — the vendor does not deny a derivation from the MBTI either — and moved the central finding to C1 alone. No coding was reopened**: all 24 flagged works were read against the corrected row before it was adopted, every one carries a derivation predicate asserted of the MBTI or of Myers and Briggs, and none rests on an unchained Jungian claim, so the counts stand where they stood. Had one rested on it, it would have been recoded and the change reported here. The second observation is that the study's stated question is about attribution while its main measure is about administration; the answer is the cross-tabulation described under Planned reporting, which is the one figure in this paper that the protocol did not name in advance and is labelled so wherever it appears. A companion entry of the same date marks two passages of the protocol as wrong where they stand — §1's claim about the peer-reviewed literature and §10's instruction that every reported figure is a lower bound — leaving the text in place and recording the correction, because a promise rewritten after the fact is no longer evidence of what was promised.

The second round returned six findings, two of which reach the protocol. S2 was being reported at the main-analysis values although its input had never been retrieved, and is now not estimable. And the defence that the cross-tabulation carried "counts and no rate" was withdrawn, because "sixteen of seventeen" is a proportion however it is printed. **That same entry lowered the exploratory attribution figure from 16 of 17 to 15 of 17**, the seventeenth work carrying C3 alone — a claim about the vendor's test's standing, which this study has no fact to weigh against.

**One post-count amendment raised a figure, and it is named here because every other one lowered a claim.** The cross-tabulation reported 15, was raised to 16, and returned to 15; the change log carries the sequence and the reason at each step. The two figures differ by definition rather than by judgement — **16 is any of C1-C3, 15 is C1 or C2, and the difference is the single work carrying C3 alone**, which is a claim about standing and not about attribution. The intermediate step departed from the rule this study followed where the protocol was silent, which is to take the reading that weakens the headline, and it is recorded as a departure at the entry that made it.

The third round returned seven findings and changed no count. Its two central ones are that limits stated in one section had not been carried into the others: the 2026 dating of the vendor's pages, which bounds what an (a) code can be said to show, and the exclusion of C3 from the attribution reading, which bounds what the union of the C flags can be called. Both are now carried throughout, and the interpretation changes they forced are entered in the departures table below.

Four further entries are dated 26 August 2026. The third and fourth are described after them; none of the four changes a coded value. A four-reader internal review — given separate angles, and one reader given the manuscript and nothing else — found that six corrections made the day before existed in the paper and not in the artefacts beside it: the published results file still asserted a defence the paper had withdrawn and still carried the reporting wording the paper departs from, and the repository's own description of the study carried both the frame and the instrument code as they read before the retreat. It also found the sentence corrected first in this section's account, and one in the Discussion. **None of it moved a count.** The entry records the two proposed corrections that were examined and refused, because refusing a correction after the counts are known needs as much of a record as making one.

The second entry of that date answers a fourth round of model-assisted review, and it corrects an arithmetic error this paper had made about its own uncertainty. A sentence added the day before claimed that the unretrieved works could not move the headline further than the interval printed beside it. Methods says in terms that these intervals carry no uncertainty from retrieval, and the arithmetic agrees: the worst case reaches 41.5%, below the interval's lower bound. That range is now reported as a worst case, and as a departure. The same round found that code (c) had been described as a silence when three of the nine works it covers describe their instrument in part, that three sentences claimed more than this study checked, that the Abstract still called the protocol fixed, and that "every reference was verified against Crossref" is false of three references carrying no DOI. **No coded value changed in any of it.**

The third entry of 26 August answers a fifth round and corrects this protocol's own attribution error. C2's row had said the flag records a lineage "the vendor disclaims". Corrected once for naming Jung, it still named descent from the published MBTI — and **the vendor denies that no more than it denies Jung**. Its pages say the two are not interchangeable, that the model is proprietary, and that Jungian cognitive functions are not incorporated, while stating that it uses "the acronym format introduced by Myers-Briggs". Being a proprietary model and being uninfluenced by the MBTI are different propositions, and only the first is asserted. **This audit had therefore attributed to a source a position the source does not hold, in the row covering 24 of its 61 coded works** — the error it exists to measure, in the same row, twice. C2 is now a count of assertions this study does not adjudicate, and the central finding is C1 alone. **No work was recoded and no count moved.**

### *Planned reporting*

Two measures are reported whatever the numbers, and neither may be dropped or moved to a supplement.

- **M1, instrument attribution.** Among E1 works in the main analysis, the share coded (a), (b) and (c), with Wilson 95% confidence intervals and the denominators stated. The name is the protocol's and is kept for correspondence with it, but it is wider than what the measure records: the codes say which instrument was administered, not how the work attributed it.
- **M2, citation and conflation.** Across all coded works, the share carrying R4 and the share carrying any pre-defined C1–C3 flag, with the same interval treatment. The protocol's name for the measure is kept for correspondence with it, but here too the name is wider than what the measure records: the union of the three flags is not a rate of conflation, because C3 alone is a claim of standing this study does not adjudicate. The three are therefore also reported separately.

What the result pattern decides is only which measure the Abstract leads with and how strongly M1 is worded. The mapping was fixed in the protocol: where the lower bound of the (a) proportion clears 0.10, the Abstract leads with M1 and states the figure directly. That row is the protocol's **P1**, and it is the branch this corpus returned, and it was evaluated by published code from the counts rather than chosen by reading them.

**One quantity is reported that the protocol did not plan, and it is labelled as such wherever it appears.** The protocol closed the list of reported figures to M1, M2 and the sensitivity arms; the conflation flags of the main analysis, split by instrument code, were added on 25 August 2026, after every count existed. The reason is that M1 answers only half of the question the Introduction asks — it records what was administered, not what the work called it — and the half it does not answer is carried by columns that were coded twice, scored and adjudicated in August, long before this cross-tabulation was computed. **An earlier version of this paragraph defended the addition on the ground that it carries "counts and no rate". That defence is withdrawn, because it is not true.** "Sixteen of seventeen" is a proportion whether or not a percentage sign is printed beside it, and a reporting rule evaded by declining to divide is not a reporting rule. The block is therefore described for what it is: **an exploratory, post hoc set of descriptive proportions**, reported without confidence intervals and without any confirmatory reading, computed by the published analysis script rather than read off by hand, and labelled unplanned wherever it appears. It is **a departure from the protocol's closed list of reported quantities**, not merely an amendment to it, and is counted among this study's departures rather than beside its planned measures.

Intervals are Wilson score intervals, which the protocol named before any count and which are the usual recommendation at this denominator, where the exact interval is markedly conservative. They are computed in Python (3.14) in the analysis script from the standard normal quantile rather than taken from a library, so that the published `requirements.txt` is sufficient to reproduce every figure. The implementation is checked in the test suite against published bounds and, where the library is available, against `statsmodels` (0.14.6) for every numerator at four denominators.

**What the intervals cover should be said plainly, because the corpus is enumerated rather than sampled.** The 27 works are not a draw from a population; they are every retrievable journal article in the intersection that administered an instrument. Within that set the share is 17 of 27 exactly, with no uncertainty at all, so an interval is only meaningful against a stated inferential target, and an earlier draft named none — it described the intervals as "the range of underlying rates compatible with these counts", which is a superpopulation claim in all but name.

**The targets are stated here, and they differ between the two measures.** For **M1** the interval is read against a hypothetical superpopulation of works the same procedure would yield: further journal articles naming the MBTI in title or abstract, containing a vendor word form in full text, retrievable without circumventing a refusal, and administering an instrument. For **M2** the target is a superpopulation of the same hypothetical kind but wider, because the denominator is wider: further works of any venue class and any gate code — including works that administered nothing — that this retrieval and coding procedure would yield beyond the 61 observed. An earlier version of this paragraph named one target and attached it to both, which left M2's two intervals without a referent, and a later one described M2's target as the observed denominator, which is not a target at all.

**The model these intervals assume is stronger than exchangeability, and naming only exchangeability was not enough.** A Wilson interval is designed for independent binomial observations and generally provides near-nominal coverage; because the binomial is discrete, its actual coverage oscillates about the nominal level rather than attaining it exactly, and no interval here should be read as covering at exactly 95%. The model it needs is that, for the indicator in question, each work is an independent Bernoulli(p) draw with the same p. Exchangeable works may still be correlated — several papers from one research group, one field, or one year — and under correlation the interval is too narrow. So the assumption stated is **i.i.d. Bernoulli(p) for each reported indicator**, and these are reported as **model-based intervals** rather than as sampling intervals.

**What they do not cover.** They carry no uncertainty from retrieval (35 works unread), from coding (two model coders of one lineage, one adjudicator), or from change over the 2015-2026 window, across which the vendor's reach grew and later works may not be exchangeable with earlier ones. None of that is testable on this corpus. The protocol requires the intervals to be reported, so they are not removed; readers who decline the model should read the counts, which are given beside every proportion and are what every claim in this paper rests on.

### *Sensitivity analyses*

Seven arms were fixed in the protocol, before any count, so that running one later could not become a way of finding a better number. All are reported whether or not they change the conclusion: **S1** adds conference and conference-abstract classes to the denominator; **S2** adds the one record whose venue metadata is absent; **S3** counts reference-list-only vendor evidence as (a); **S4** restricts to OpenAlex; **S5** widens the word-form variant; **S6** excludes works whose retrieved text is a conference abstract; **S7** reads the conflation flags narrowly, keeping only works that name the vendor's test or site.

### *Calibration*

Three records were identified before coding as documents the pipeline had to contain, with their expected codes written from verbatim already verified against the sources, and were given to the coders as calibration. **Two of the three fall outside the main analysis** — one is a conference abstract and therefore outside the journal-article class, and one administers nothing and so takes no instrument code. That is stated here, in Methods, rather than discovered in Discussion, and it is the reason the citation-role and conflation steps are coded on the whole corpus rather than on the subset that administered something.

Two consequences of that design are stated here rather than left for a reader to notice. **The calibration is not an independent check on the coders.** They were given the expected codes before coding, so reproducing them shows that the pipeline retrieved these records and put them through the protocol; it does not show that the coders read correctly, and no reliability claim rests on it. And **the third record is inside the main analysis and inside its numerator**: it is a journal article, it administered the vendor's test, and it is therefore one of the works coded (a). Its instrument code was written in advance and handed to both coders. One of the seventeen (a) works in the headline is a work whose answer was supplied.

### *Departures from the original plan*

The protocol was written before any work was classified and its change log has carried thirteen dated entries since. Saying that the reporting "was fixed in advance" is therefore true of the first version and not of the document as it now stands, and the difference is set out here rather than left to be reconstructed from a change log. Each row gives what was promised, what was done, what was known at the time, and what it moved. **The table lists fourteen distinct departures and the change log carries thirteen dated entries; the two do not correspond one to one**, because several departures were recorded together within a single dated entry.

| Promised | Done | Known at the time | Effect on the numbers |
|---|---|---|---|
| §5 and §6 coded on all 99 works | Coded on the 61 with retrievable full text | No coding had started | None; the wording contradicted §1 |
| §5 as first written on what a report of another study's use of the vendor carries | Clarified as R6, mention only, so that a work does not accumulate another study's practice into its own row | Two calibration records coded; no proportion computed | None; no expected code changed, and one coder's record of that work was discarded and coded again |
| Conflation rules as first written | Twelve, then three further questions answered; C flags re-coded twice on all 61 works | Both κ values were known; no proportion had been computed | C2's first-round κ was 0.52; the reported 0.859 describes the final round |
| §10's reporting table to determine a claim for every result | Four gaps in it settled | All coding complete; no proportion computed | None on this corpus: $n_1$ = 27, no tie, same branch |
| "Every figure is a lower bound" (§10) | Counts reported as lower bounds; the direction of bias in the proportions reported as unknown | Every count known | No figure moved; the claim attached to them weakened |
| §1: the claim is about "the peer-reviewed literature" | Claim restated as the journal-article literature | Every count known | None; the scope claim narrowed |
| §12: only M1, M2 and the arms are reported | Table 2 added — the conflation flags of the main analysis by instrument code | Every count known | Adds an exploratory result; the planned measures are unchanged |

| Promised | Done | Known at the time | Effect on the numbers |
|---|---|---|---|
| §8: the venue-less record coded in full so S2 runs on real codes | Its full text was never retrieved; S2 reported as not estimable | Retrieval was complete | S2 has no result, where an earlier draft reported it as unchanged |
| §6's C2 row, which named lineages as ones "the vendor disclaims" | Corrected twice: first to drop Jung, whom the vendor claims of itself; then, the vendor denying no derivation from the MBTI either, C2 was redefined as a count of asserted derivations that this study does not adjudicate, and the central finding narrowed to C1 alone | Every count known | None; C2 stays at 24 and any C1–C3 at 44. The headline moves from 15 of 17 to **13 of 17** |
| §6's `states_distinction` read as its name suggests | Described as recording that a difference was noted somewhere; on the reading that asks whether the work separated the instruments, the count is 2 of 3 | Every count known | None; the field is reported unchanged and its description corrected |
| §10's M2 read as a conflation rate | Reported as the share carrying any pre-defined C1–C3 flag; C3 reported separately and excluded from the attribution reading | Every count known | None; 44/61 unchanged, the reading of it narrowed |
| §3's (a): "The instrument administered was the 16Personalities test / NERIS Type Explorer" | Read as "vendor-hosted, and no published MBTI form identifiable from the work"; non-identity at the date each paper was written is not claimed | Every count known | None; no code moved, the claim attached to (a) narrowed |
| §10's P1 claim: "X% (95% CI …) administered an instrument that is not the MBTI" | Reported as a vendor-hosted test from which no published MBTI form was identifiable, the vendor's pages having been retrieved only in 2026 | Every count known | None; the branch is still P1 and the share is still 63.0%, the claim attached to it narrowed |
| §12: only M1, M2 and the arms carry a proportion | A worst-case range for M1 over the unretrieved journal articles (41.5%–75.6%) is reported in Limitations 2 | Every count known | Adds two exploratory proportions; no planned figure moved. An earlier sentence claimed the interval already covered this, which is false — 41.5% falls outside it |

The Table 2 row and the S2 row are the ones a reader should weigh most. **Table 2 is a departure, not merely an amendment**: the protocol had closed its list of reported quantities, and this adds to it after every count was known. It is kept because it answers the question the Introduction asks and the planned measures do not, and it is labelled exploratory everywhere it appears. **A study built on pre-commitment has to count its departures, not describe them as edits**, which is why they are tabulated here and not only in the protocol's log.

### *Data and code*

The corpus, the retrieval log, the protocol, the per-work classification, the agreement table, the author's rulings, the analysis script and its tests are published (see Data Availability). Full texts and third-party documents are not redistributed.

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## Results

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### *Corpus*

The two queries returned 118 records resolving to 99 distinct works: 58 journal articles, 13 conference papers, 10 repository deposits, 7 preprints, 6 theses, 2 book chapters, 1 conference abstract, 1 non-scholarly item and 1 record whose venue metadata is absent. Full text was retrieved for 64 works and coded for 61, including 42 of the 58 journal articles (72.4%).

Of the 61 coded works, the gate placed **34 in E1** (respondents completed an instrument for the study), 13 in E2 (type labels taken from existing data), 6 in E3 (administered to a language model or other artificial agent) and 8 in E4 (no type data). The 34 E1 works are distributed across venue classes as 27 journal articles, 3 conference papers, 2 repository deposits, 1 book chapter and 1 thesis. **The main analysis therefore rests on $n_1$ = 27.**

### *M1 — what was administered*

Table 1 and Figure 2 give the instrument distribution among the 27 E1 works in the main analysis.

**Table 1.** Instrument administered, among E1 works in the main analysis ($n_1$ = 27), with model-based Wilson 95% intervals (see Methods). (The protocol's name for this measure is *M1, instrument attribution*; the codes record what was administered, not how the work attributed it.)

| Instrument code | Works | Share | Model-based Wilson 95% interval |
|---|---|---|---|
| **(a)** a vendor-hosted test; no published MBTI form identifiable | **17** | **63.0%** | 44.2–78.5% |
| **(b)** a published MBTI form | **1** | **3.7%** | 0.7–18.3% |
| **(c)** other or insufficiently identified; no published MBTI form identifiable | **9** | **33.3%** | 18.6–52.2% |
| **Total** | **27** | 100% | — |

**Of the 27 journal articles in this corpus that name the MBTI and administered an instrument, 17 (63.0%, model-based Wilson 95% interval 44.2–78.5) administered a vendor-hosted test from which no published MBTI form was identifiable.** One work in the main analysis — 3.7% of it — administered a published MBTI form. Thirty-five works in the frame could not be retrieved, and that count belongs beside every proportion above. **This is a distribution of instruments administered, and on its own it is not a rate of misattribution**: a work enters the corpus by a string match, not by claiming to have administered the MBTI. What each work called the instrument it used is reported below.

Across all 61 coded works rather than the main analysis alone, the 34 E1 works divide as 22 (a), 1 (b) and 11 (c); the additional (a) works sit in two conference papers, a repository deposit, a book chapter and a thesis.

The 11 works coded (c) divide by the shape of the gap as follows: 4 stated only that "the MBTI" was administered, with no form, publisher, version or URL; 2 used items the authors wrote themselves; 1 an unnamed online test; 1 a translated questionnaire of unstated provenance; 1 cited a vendor page in its reference list and nowhere else; and 2 fit none of the protocol's sub-labels and were recorded in free text. **Sub-labels do not affect the main code**, and no claim here rests on them.

### *What the works called what they administered — an unplanned cross-tabulation*

**Nothing in this subsection was planned**, and it is set apart for that reason. The protocol closed its list of reported figures before the first count; this cross-tabulation was added on 25 August 2026, after all of them existed, for the reason given in Methods. It is reported as exploratory, post hoc descriptive proportions against a stated denominator, without model-based intervals and without any confirmatory reading. The columns it reads — the conflation flags and `states_distinction` — were coded independently by both coders, scored for agreement and adjudicated in August, so no work was read again to produce it.

Table 2 splits the 27 works of the main analysis by instrument code against what each work says its instrument *is*.

**Table 2.** Pre-defined C flags among the works of the main analysis, by instrument code. Exploratory, post hoc descriptive proportions; no model-based intervals.

| **Pre-defined C flag** | **(a) vendor-hosted, no MBTI form identifiable** | **(b) published MBTI form** | **(c) other or insufficiently identified** |
|---|---|---|---|
| **Works** | 17 | 1 | 9 |
| C1, named as one instrument | 13 | 0 | 4 |
| C2, given the MBTI's provenance | 10 | 0 | 2 |
| C3, claimed the standing of a published instrument | 7 | 0 | 1 |
| C1 or C2 | 15 | 0 | 5 |
| **Any of C1–C3** | **16** | **0** | **5** |
| C0, none of C1–C3 | 1 | 1 | 4 |

| Pre-defined C flag | (a) vendor-hosted, no MBTI form identifiable | (b) published MBTI form | (c) other or insufficiently identified |
|---|---|---|---|
| `states_distinction`, noted some difference between the two | 3 | 0 | 0 |

The counts separate, and they should be read separately rather than pooled, because the three flags do not carry the same weight of evidence:

- **13 of the 17 described the vendor's test as the MBTI** (C1) — **the central finding**, and the only one of the three that records a claim the vendor addresses directly.
- **15 of the 17 either did that or asserted that the vendor's test derives from the MBTI** (C1 or C2) — the broader pre-defined reading. The vendor does not deny such a derivation, so this line counts assertions and does not evaluate them.
- **16 of the 17 carried at least one of the three pre-defined C flags** (C1, C2 or C3).

**The first line is the one that stands on its own.** The vendor states plainly that its test is not the MBTI, so a work calling it the MBTI attributes to it an identity its operator denies. The second line adds works asserting a derivation the vendor neither claims nor denies, so it is reported as a count of assertions. The seventeenth work in the third line carries C3 alone: it claimed for the vendor's test the standing of a published instrument — "a validated instrument", or the equivalent — without asserting either identity with the MBTI or descent from it. That is a claim about standing, and this study establishes no fact about the vendor's test's validity against which such a claim could be called false. **It is reported in the row because the protocol defined it before any count, and it is excluded from the attribution reading for the same reason the protocol drew C3 as a separate flag.** Three carry the protocol's `states_distinction` field, and **the field is weaker than its name**: it records that a work noted some difference between the two, not that the work attributed its instrument correctly. Their quotations differ in kind, and one of the three does not survive reading. One writes that its instrument is "grounded in the Big Five framework" despite reporting "results in the familiar MBTI format". A second, writing in Indonesian, records that "the theoretical basis of the two platforms is not the same" — the official MBTI Jungian, 16Personalities from the Big Five (translated; the original is in the classification file). **The third is not the same kind of statement at all**: it writes of "16 types identical to the Myers-Briggs test", noting only that the vendor appends a fifth letter to a type code it treats as the MBTI's. That is a difference in format asserted alongside an identity of content, and the work carries C1. Two of the three carry C1 in their own right, the first and the third. **On the reading that matters — did the work separate the two instruments — the count is two, not three**, and the field's own name should not be read as making the stronger claim. The field is reported unchanged because it was coded, scored and adjudicated before any of this was computed; what is corrected is the description of it. **The flags are set per statement, so a work can appear in more than one row**, and the reader who wants to separate these cases should read the located quotations in the published classification file rather than the totals here.

**One work — one of the seventeen — administered the vendor's test and carried none of the three pre-defined C flags.** The single work that administered a published MBTI form carried none either. Every column of Table 2 closes on its flagged-plus-unflagged total: sixteen and one for (a), nought and one for (b), five and four for (c).

### *M2 — how the vendor is cited and described*

Table 3 gives the two planned citation and flag measures across all 61 coded works.

**Table 3.** The two reported measures across all coded works (n = 61), with model-based Wilson 95% intervals (see Methods).

| Measure | Works | Share | Model-based Wilson 95% interval |
|---|---|---|---|
| **Any pre-defined C1–C3 flag** | **44** | **72.1%** | 59.8–81.8% |
| **R4 (vendor cited as psychometric evidence)** | **2** | **3.3%** | 0.9–11.2% |

**That 44 is a union of three flags, and it is not a conflation rate.** C1 attributes to the vendor's test an identity the vendor denies; C2 records an asserted derivation the vendor neither claims nor denies; C3 claims for it the standing of a published instrument, and this study establishes no fact about that test's validity against which such a claim could be called false. Read separately across the 61 coded works, **37 carried an identity claim (C1), 24 a provenance claim (C2), and 15 a claim of standing (C3) whose truth is not judged here.** The union is reported under M2 because the protocol fixed it before any count existed, and it is named for what it is: the share carrying at least one pre-defined C flag.

The individual flags are shown in Figure 3 and reported here as counts, because the protocol assigns a rate only to the two measures above: for citation role, R1 instrument 28, R2 theory 21, R3 norms 3, R4 psychometrics 2, R5 data source 4, R6 mention only 8, R7 object of study 6; for conflation, C0 none 17, C1 identity 37, C2 provenance 24, C3 authority 15. Under the narrow reading the counts are C1 36, C2 24, C3 15 — a difference of a single work.

Seven works are flagged in `states_distinction`, which records that a work noted some difference between the two rather than that it separated them correctly; a work can appear both here and in the conflation counts, because a correct sentence does not undo a conflating one. Five works applied the name "MBTI" to a third look-alike instrument altogether. **That shape is recorded and described and is never rated**: the corpus is built from 16Personalities word forms, so works conflating other look-alikes enter it only incidentally and no proportion over them would have a denominator. Four works were flagged by both coders as having a retrieved text that is an abstract rather than an article, despite their metadata.

### *Agreement*

Table 4 gives inter-coder agreement by code.

**Table 4.** Cohen's κ by code. The conflation flags describe the final coding round only (see Methods).

| Code | n | Cohen's κ |
|---|---|---|
| Gate (E1–E4) | 61 | 0.974 |
| **Instrument (a)/(b)/(c), works both coders placed in E1** | **34** | **1.000** |
| R1 / R2 / R3 | 61 | 0.967 / 0.852 / 0.849 |
| R4 / R5 / R6 / R7 | 61 | 0.792 / 0.849 / 0.796 / 1.000 |
| C0 / C1 / C2 / C3 | 61 | 1.000 / 0.965 / 0.859 / 0.858 |
| Narrow C1 / C2 / C3 | 61 | 0.966 / 0.859 / 0.858 |
| `states_distinction` | 61 | 0.914 |
| `text_is_abstract` | 61 | 1.000 |

The instrument code — the one that carries M1 — agreed on every work both coders placed in E1, and was never re-coded at any stage. **These coefficients did not move when the author's rulings were applied**, and are not expected to: kappa describes the coders' raw output, and adjudication sits above it.

### *Sensitivity analyses*

Table 5 and Figure 4 give the two measures under the main analysis and each planned arm.

**Table 5.** Planned sensitivity analyses, with model-based Wilson 95% intervals (see Methods). Five arms could be run; S2 and S5 could not, for the reasons given below.

| Arm | M1 (a) | M2, any pre-defined C1–C3 flag |
|---|---|---|
| **Main analysis** | 17/27 = 63.0% (44.2–78.5) | 44/61 = 72.1% (59.8–81.8) |
| **S1** conference classes added | 19/30 = 63.3% (45.5–78.1) | unchanged |
| **S2** venue-less record added | not estimable | not estimable |
| **S3** reference-list-only counted as (a) | unchanged | unchanged |
| **S4** OpenAlex only | unchanged | unchanged |
| **S5** widened word form | bound: 17–18/28 | bound: 44–45/62 |
| **S6** abstract-texts excluded | 16/26 = 61.5% (42.5–77.6) | 42/57 = 73.7% (61.0–83.4) |
| **S7** narrow conflation reading | unchanged | 43/61 = 70.5% (58.1–80.4) |

Four arms warrant a sentence each. Neither S2 nor S5 yields an estimate, and the Abstract reports both that way, but they fail differently: S2 has no result because its input was never obtained, while S5 is reported as a bound on an effect it cannot recount. S3 and S4 return the main-analysis figures for reasons that are not the same either.

**S2 did not run, and is reported as not estimable rather than as unchanged.** The protocol undertook to code the venue-less record in full so that this arm would run on real codes rather than on a hole. Its full text was never retrieved, so it is `unobtainable` and carries no code to add. An earlier version of this table gave S2 the main-analysis figures under the heading "unchanged", and the figure plotted it beside the arms that ran, with a point estimate and an interval. That asserted a result the arm never produced. **An arm whose input was never obtained has no result, which is not the same as a result equal to the main analysis.**

**S3 moves nothing because the shape it re-reads sits outside the denominator.** Exactly one coded work carries reference-list-only vendor evidence, and it is a repository deposit, not a journal article. The conservative boundary rule and its permissive alternative therefore produce the same main analysis; the arm is reported because it was fixed in advance, not because it was informative.

**S4 is an identity.** All 61 coded works are in OpenAlex, so restricting to that source removes nothing. This is the measured form of the statement made in Methods: Europe PMC confirms the frame and does not extend it.

**S5 is a bound rather than a recount.** The widened word form takes the OpenAlex intersection from 108 records to 109 and leaves Europe PMC at 10. The added record is not in the corpus: it was never retrieved and carries no code. The arm therefore bounds the effect at one record — at most one work against a frame of 99. The record's venue class and gate status are unknown, because it was never retrieved, and the two measures have different denominators, so the arm bounds them under different conditions. **If the record could not be retrieved and coded at all**, neither measure moves. **If it could be coded but is not a journal article or not an E1 work**, M1 does not move and M2 lies between 44 and 45 of 62, its denominator being every coded work of any venue class and any gate code. **If it could be coded and is both**, M1 lies between 17 and 18 of 28 and M2 between 44 and 45 of 62. Those are the bounds the arm asserts, and they are conditional; no code is invented for the record, and the arm is not dropped in silence.

### *Calibration*

All three calibration records reproduced every code the protocol had pre-judged for them. One is a journal article that administered the vendor's test and cited the vendor's own page for reliability coefficients; it is in the main analysis and coded (a) with R1 and R4. One is a conference abstract that identifies the two instruments as one and, having done so, asserts the published MBTI's lineage of the vendor's test — the chain §6's coding rule requires, rather than a bare Jungian ancestry, which the vendor claims for itself and which sets no flag; it is outside the main analysis on venue class. One is a journal article that administers nothing, mentions the vendor only in describing studies it cites, and calls the vendor's test "a popular MBTI questionnaire"; it takes no instrument code and is therefore outside M1 despite being a journal article. **Two of the three fall outside the main analysis**, as fixed in advance.

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## Discussion

Within an analytic corpus conditioned on programmatic retrievability and built from the intersection of an MBTI mention and a 16Personalities word form, journal articles that administered an instrument mostly administered a test hosted by the vendor, with no published MBTI form identifiable from the paper. Adding the works whose description does not identify a published MBTI form, **26 of the 27 works in the main analysis did not demonstrate that they administered the MBTI** — 17 because what they administered was the vendor-hosted test, 9 because what they describe does not identify one. One work did.

That is a statement about instruments, not about attributions, and the two must be kept apart even though the corpus makes them easy to merge. The finding that bears on attribution is the exploratory cross-tabulation, and it has to be read in layers. Of the 17 works that administered the vendor's test, **13 called it the MBTI** and **15 either called it the MBTI or gave it the MBTI's provenance**. A sixteenth claimed for it the standing of a published instrument without asserting either, which is a different kind of claim and is not counted among those fifteen. **Thirteen of seventeen described the vendor-hosted test as the MBTI, contrary to the vendor's 2026 statement that the two are not the same; seventeen of twenty-seven is the distribution of instruments administered.** Fifteen of seventeen is the broader pre-defined reading, which also counts a work asserting that the vendor's test derives from the MBTI — a claim the vendor does not address, so the study records it without judging it. They are given as fractions, and not as "the first" and "the second", because a reader who takes one for the other will overstate this study — an earlier draft of this sentence made that slip, a later one pooled all three flags into the same figure, and a later one still called the fifteen a misattribution, which asserts a non-identity at the date each paper was written that this study did not establish.

Three features of that result deserve separate treatment, because they are three different problems.

**The substitution.** The word names a relation between what was administered and what was named, and carries no claim about the vendor's product specification at any date: seventeen works administered a test whose operator states, on pages retrieved in 2026, that it is not the MBTI, and thirteen of them described it as the MBTI. Nothing here establishes intent, and the most parsimonious reading requires none: the vendor's test is free, immediately accessible, and returns the same four-letter output. A researcher looking for "the MBTI" online may well arrive at it — though this study measured neither search rankings nor the paths these authors took, so that sentence is a hypothesis the design cannot test. What follows is not a moral failure but a measurement one — the type distributions reported in these papers were produced by a test its operator now states is not the MBTI, and a reader pooling them with results from a published MBTI form is pooling results whose measurement equivalence this study did not assess.

It bears emphasis that the vendor's own dating forecloses the sharpest version of the criticism. The machine-readable page quoted in the Introduction carries the stamp "Last updated: June 11, 2026". A paper published before that date could not have consulted it, so it cannot be used to argue that any particular author should have known. Whether the distinction was discoverable by other means at the time, and by what means, is not something this study examined; what is certain is that this particular statement of it was not available at the time most of these papers were written.

**A second limit of the same kind runs the other way, and is not resolved here.** Both vendor pages quoted in this paper were retrieved in 2026, and they describe the instrument as it stands in 2026. Nothing in this study establishes that the test offered in, say, 2016 was the same instrument on the same scales under the same terms: that would require archived copies of the pages as they then stood, the terms of service of the day, and a version history of the items and scales, none of which was collected. **The claim the coding actually needs is weaker still, and this paper does not make the stronger one.** What each (a) code records is that the work administered a test hosted by the vendor and that no published MBTI form is identifiable from the work. Whether the vendor's product was, at every date from 2015, distinct from the MBTI is not established by anything here: the only evidence offered is dated 2026. Readers should take (a) as "vendor-hosted, and no published MBTI form identifiable from the work" and not as a finding about the instrument's specification at the time each study ran. But no statement here should be read as certifying the instrument's specification at any date before retrieval.

**The partial descriptions.** Nine works administered something from which no published MBTI form is identifiable. That is not the same as saying nothing: their descriptions vary, and include author-constructed items, a translated questionnaire of unstated provenance, an unnamed online test, and works that named "the MBTI" and no form, publisher, version or URL. It would be convenient to call this non-compliance with established reporting guidance, and that is not available: a three-part review of survey-reporting practice found that fewer than 7% of 165 medical journals gave authors any guidance on survey research, that the four published checklists it identified were unvalidated, and that key criteria were poorly reported in the 117 survey papers it examined — concluding that "there is limited guidance and no consensus regarding the optimal reporting of survey research" (Bennett et al. 2011). The incomplete specification is therefore not a departure from a settled norm in the reporting of survey research, which is the literature Bennett and colleagues examined. Whether psychological measurement more broadly has such a norm is a separate question this study did not investigate, and the claim here is limited to the one the cited evidence supports. That makes the reporting gap more consequential rather than less. A paper that names no form, publisher, version or URL cannot be replicated with respect to its central measurement, and cannot be audited even by a reader who suspects a substitution. In a corpus of this shape an incomplete specification is not neutral: it is the state in which a substitution would be invisible.

**The flags in prose.** Nearly three-quarters of all coded works — including reviews, position pieces and machine-learning papers that administered nothing to anybody — carried at least one pre-defined C flag: 37 treated the vendor's test and the MBTI as a single instrument, 24 attributed to it the published instrument's provenance, and 15 claimed for it the standing of a published instrument. **The third of those is not a conflation and the union should not be quoted as a conflation rate**; only the first records a claim the vendor addresses; the derivation and the claim of standing are recorded as assertions this study does not adjudicate. The provenance flag is narrower than its name suggests and is worth restating here, because the audit would otherwise commit the error it measures: it does not fire on a work that traces the vendor's test to Jung, which is what the vendor says of itself. Every work carrying it asserts descent from the MBTI or from Myers and Briggs, and the eleven that reach Jung do so only after having already named the vendor's test as the MBTI. This is where the instrument codes cannot carry the argument alone: a review that never touches a respondent still reproduces the conflation when it calls the vendor's test "a popular MBTI questionnaire", and a work like that is invisible to any analysis of what was administered. Two of the three calibration records are exactly that shape.

One mechanism consistent with this pattern — not compared against others, because this design cannot compare them — is the one the citation-accuracy literature already describes: a claim propagates by being copied rather than by being checked. Stang and colleagues' case study is the closest analogue — a document cited overwhelmingly as saying the opposite of what it says, by authors who evidently read the citation rather than the source. Our failure is not identical, and the wording should not blur them: quotation error concerns whether a source supports the claim attached to it, whereas this study concerns whether the instrument named is the instrument used. They are neighbours, not the same thing, and no rate reported here should be compared against a quotation-error rate.

**What this study does not show.** It does not show *why* the pattern occurs: the design codes what each work says, and traces no citation lineage, so the mechanism suggested above is a reading of this result against prior work rather than a test of it. It does not show that the vendor's test is invalid, and nothing here bears on that question. It does not show that the MBTI is well supported; that question is settled elsewhere and is not ours, though the long-standing critique of the instrument's psychometrics is part of why the gap between the instrument administered and the instrument named matters (Pittenger 2005). It does not show that independent psychometric evidence for the vendor's test is unavailable: a confirmatory factor analysis of the vendor's scale on 1,067 respondents was published in 2020 (Makwana & Dave 2020), five years before the calibration record that sourced reliability coefficients to the vendor's own product page. What this study did **not** do is assess how much such evidence exists or how good it is; what it can report is one paper's discoverability. That paper has six citations (OpenAlex, retrieved 22 August 2026) and its venue is not in the Directory of Open Access Journals (DOAJ). Its publisher describes it on the article's first page as Scopus-indexed; Elsevier's own Scopus source list, July 2026 edition, records the journal as "Inactive" with coverage "2019-2020" and flags it "Discontinued by Scopus", and the same workbook's discontinued-titles sheet gives its final covered issue as volume 11, issue 6 of 2020 — while the paper appeared in volume 11, issue 9. Discoverability, not soundness, is the point; the paper's merits are not at issue here, and the venue was indexed for a period, so it is not correct to say it was never in Scopus. The workbook is a third-party document and is cited as one, with its retrieval URL, date and checksum in the provenance record.

**The two coders.** The agreement figures in this paper describe two tiers of one vendor's model line reading the same protocol, and they should be read as measuring the stability of one lineage's reading. The instrument code, which carries the headline, agreed on every eligible work and was never re-coded — but two coders drawn from one lineage agreeing perfectly is weaker evidence than two independent readers agreeing perfectly, and the difference is not quantifiable from these data. What is checkable is the coding itself: every code carries a located verbatim quotation, and the classification file, the protocol and the rulings are published so that any reader can disagree with a specific work rather than with a coefficient. One asymmetry is worth reporting for the same reason: across three rounds, proposals to resolve splits came disproportionately from the second coder, and because the proposing agent and that coder are the same model, whether this reflects a better reading or a shared bias cannot be determined from this design.

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## Limitations

1. **The corpus is enriched by construction, the headline rate is not a population rate, and it is not a rate of misattribution.** Works enter only if they mention the MBTI *and* contain a vendor word form. That intersection selects for exactly the papers most likely to have used the vendor's test. The 63.0% is a rate within the intersection and is not an estimate of how often papers reporting MBTI results in general administered a vendor-hosted test; the corresponding frames — 3,105 OpenAlex works and 166 Europe PMC records from 2015 — are given in Methods so the reader can see the scale of the gap between the two quantities. A second gap runs in a different direction and is easier to miss: **entering on a string match is not the same as claiming to have administered the MBTI**, so M1 measures which instrument was used and not whether the work named it correctly. In fact 3 of the 17 works coded (a) noted some difference between the two in their own text, and on the reading that matters — did the work separate the instruments — the count is two. The attribution question is answered instead by the cross-tabulation in Results, which was added after the counts existed and is reported as exploratory, post hoc descriptive proportions without model-based intervals.
2. **The analytic corpus was conditioned on programmatic retrievability by declaration — not on open access — and the observed positive counts are lower bounds while the proportions are not.** Thirty-five of 99 works could not be retrieved; of the fourteen missing journal articles, eight are publisher refusals and two are TLS failures, as Methods sets out. A work whose text cannot be read cannot be coded at all, so every count of a flag or a code can only rise as coverage improves. **The proportions have no such guarantee**: if the missing works are less often (a) than those retrieved, 63.0% falls; if more often, it rises. A paywalled paper is not obviously more or less likely to name its instrument, which is exactly why the direction of the bias in the proportions is unknown rather than conservative. The protocol's §10 fixed the wording "every figure is a lower bound" before any count existed, and that instruction is false: it is true of the counts and false of the rates. **Reporting the rates without a lower-bound claim is therefore a departure from a pre-committed reporting instruction, and is recorded as one** — §10's text is left standing as the record of what was promised, with the error marked there and logged in the protocol's change log. **The interval does not cover this, and an earlier version of this paragraph said it did.** Methods states that the model-based intervals carry no uncertainty from retrieval, and the arithmetic bears

that out: if all fourteen unretrieved journal articles were eligible E1 works, the observed share could run from **17/41 (41.5%) to 31/41 (75.6%)**, and the lower end of that range falls outside the interval printed beside the headline. **That range is a worst case, not a confidence interval**, and it models no direction of retrieval bias — the fourteen works' gate status is unknown, so the denominator is as uncertain as the numerator. It is reported because the sentence it replaces asserted the opposite, and because a reader is owed the arithmetic rather than an assurance.

3. **$n_1$ = 27, the interval is wide, and it rests on an assumption that cannot be checked here.** The main analysis rests on 27 works and the headline interval runs from 44.2% to 78.5%; the point estimate should not be quoted without it. The single (b) work carries an interval from 0.7% to 18.3%, and no claim should rest on the observation that only one work administered a published form. Because the corpus is enumerated rather than sampled, the share within it is exact and the interval is meaningful only against the inferential target stated in Methods — the works this retrieval and coding procedure would yield beyond the ones observed. **That target requires more than exchangeability — the intervals assume that each work is an independent Bernoulli(p) draw with a common p for the indicator in question — and neither assumption can be tested on this corpus, while the frame gives reason to doubt both**, the window running from 2015 across a period in which the vendor's reach grew. A reader who declines the target should read the counts, which are reported beside every proportion.

4. **Every human judgment in the study was made by one unblinded person, and the coders are two tiers of one model line.** All 31 rulings on contested items and all 35 decisions to leave an uncertain work unchanged were made by the author, who also framed the question and wrote the protocol; there was no second human reader and no blinding. **Nor can that check now be recovered on this corpus**: the only person who could perform it has already read every contested item and every flagged-uncertain work, so a spot-check by this author blind to the coders' output is no longer available at any cost. What was a missing check at the design stage is a permanent one. The two model coders measure within-lineage stability rather than the convergence of independent judgments, and are not comparable to inter-rater reliability between humans; the proposal asymmetry noted in the Discussion cannot be separated from a shared bias by this design. The mitigation is not independence but inspectability — every code carries a located quotation and every ruling its reasoning, both published. **Language compounds this.** No language or geographic restriction was applied, and at least 16 of the 99 works carry a title outside plain ASCII, with an unknown further number in ASCII but not in English. Coders quoted the original and glossed it, so a reader can check any code against the work's own words — but nothing here establishes that a Bulgarian or Uzbek full text is coded as reliably as an English one, and the agreement figures are pooled across languages rather than reported by language, which a corpus of this size cannot support.

5. **The protocol's change log carries thirteen dated entries, twelve after coding began, two of them costly and eight made after every count existed.** Two rounds discarded and re-coded flags on all 61 works. Every entry is dated and records what had been seen, and those that could raise a rate carry sensitivity arms — but a protocol amended this often describes a coding task that was under-specified when it began, and the history is reported as a property of the instrument rather than as a defect repaired. One consequence reaches the results: the conflation kappas describe the final round only, after rules were written to resolve the disagreements that had depressed them. **The eight entries dated 25 and 26 August 2026 break the pattern of the five before them**, which all preceded the first proportion: they were written with every count known, and the first of them added the cross-tabulation reported above. No coding was reopened and no planned figure moved, and the addition is labelled exploratory and carries no model-based interval — but a quantity chosen after the numbers are visible cannot carry the guarantee the rest of the protocol was written to provide, and it is not claimed to.

6. **Venue class is metadata, and metadata can be wrong about what a document is.** Four retrieved texts are abstracts rather than articles, at least one filed as a journal article in a conference supplement. Records were not moved between classes after being read, because that is precisely the boundary shift the planned reporting forbids; S6 measures the effect instead of assuming it, and moves the headline by 1.5 percentage points.

7. **Two planned arms could not run, and the calibration is not an independent check.** The protocol undertook to code the venue-less record in full so that S2 would run on real codes; its full text was never retrieved, so the arm is empty rather than informative. S5's added record was likewise never retrieved and is reported as a bound. Separately, the three calibration records were given to the coders with their expected codes, so their reproduction shows the pipeline reached them and does not evidence coding accuracy — and one of the three is inside the main analysis and inside the (a) numerator.

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**Calibration records (also members of the corpus):**

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## Ethical considerations

This study analysed published scholarly works and publicly available bibliographic metadata. No human participants were involved, no individual-level data were accessed, and no personal information was collected or processed. Under the Japanese Ethical Guidelines for Medical and Biological Research Involving Human Subjects (2021 revision), research of this kind does not require ethics committee review, and no institutional review board approval or waiver was sought. The study was conducted in accordance with the principles of the Declaration of Helsinki where applicable to research that does not involve human subjects.

Copyrighted full texts obtained for coding are not redistributed. The published repository contains the coding of each work, the located quotations on which each code rests, and the retrieval log, but not the documents themselves. Where a publisher refused programmatic access, the refusal was recorded and the work was left uncoded rather than obtained by other means.

## Acknowledgments

Large language models were used substantively in this work and their role is described here rather than in general terms. Two models supplied by Anthropic — `claude-sonnet-5` and `claude-opus-5` — served as the two independent coders described in Methods, each classifying every work against the published protocol without sight of the other's output; the instructions given to them are published with the data. The same model line assisted with corpus assembly, code, and drafting. All rulings on contested items were made by the author against the full text. The rounds of review described in Methods were conducted by a large language model — not by a human referee, and not by either of the two models that served as coders — given only the typeset manuscript and a written brief, without the protocol, the classification file or the repository. Two of this paper's central retreats originate in those reviews: the scope of what code (a) is said to show, and the reading of the union of the C flags. Two internal reviews were also run on the same model line: three instances reading the manuscript against the protocol, which produced the arithmetic correction described in Methods, and a later four-reader review on separate briefs — the revision's consistency, a recomputation of every count the paper makes about itself, a reading of the manuscript with nothing else supplied, and the code — whose findings are recorded in the protocol's change log. **No human peer review has been conducted on this manuscript.** The author is responsible for the entire content, including every figure and every reference. All eleven DOI-bearing scholarly references in this manuscript were resolved against Crossref; the remaining three — two vendor web

pages and a publisher's source list — carry no DOI and are cited with their retrieval dates; two author attributions that had entered the working literature notes were found to be incorrect and corrected before drafting.

## Author Contributions (Contributor Roles Taxonomy, CRediT)

Mizuki Shirai: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization, Supervision, Project administration.

## Conflict of Interest

The author declares no competing interests, per the International Committee of Medical Journal Editors (ICMJE) guidelines. The author has no financial or personal relationship with The Myers-Briggs Company, NERIS Analytics Limited, or any provider of personality assessment.


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## Data Availability

The corpus, the query log with retrieval dates, the coding protocol including its full amendment history, the per-work classification with located quotations, the inter-coder agreement table, the author's rulings, the record of works read without change, the analysis script, its test suite, the computed results, and two verification records — one resolving all eleven DOI-bearing references against Crossref and one matching every quotation in this paper against the text it is attributed to — are available at https://github.com/rehabilitation-collaboration/mbti-attribution-audit. Bibliographic metadata derives from OpenAlex and Europe PMC, both openly licensed; the queries and their retrieval dates are in `data/query_log.json`. Full texts of the coded works are not redistributed, and third-party documents obtained during the study are held only locally, with their provenance — source URL, retrieval date, method and SHA-256 checksum — recorded in the repository.

## Tables

**Table 1.** Instrument administered, among E1 works in the main analysis (in Results).

**Table 2.** Pre-defined C flags among the works of the main analysis, by instrument code (in Results).

**Table 3.** The two reported measures across all coded works (in Results).

**Table 4.** Inter-coder agreement by code (in Results).

**Table 5.** Sensitivity analyses (in Results).

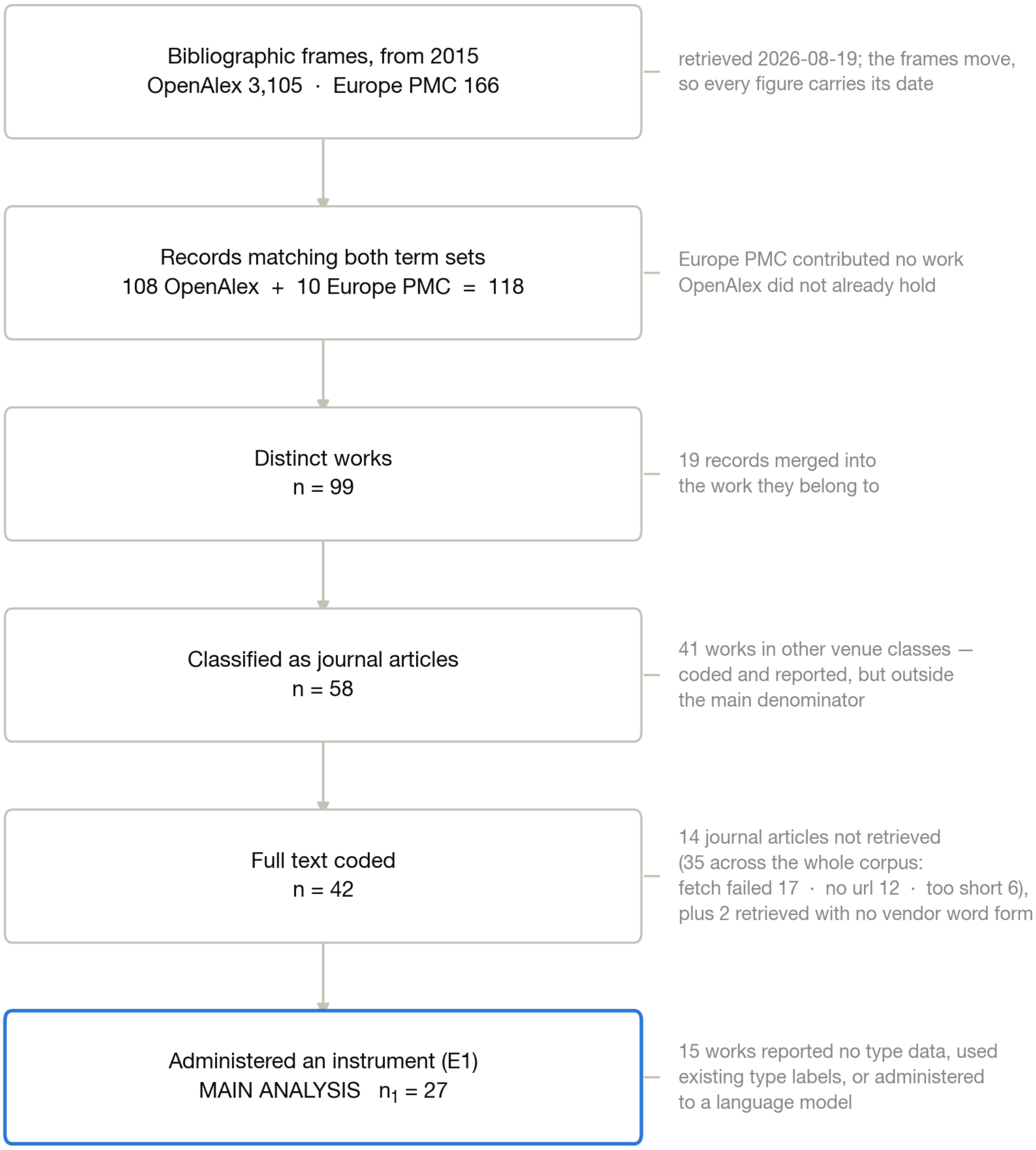


**Figure 1.** Flow of works from the two bibliographic frames to the main analysis. Frames (OpenAlex 3,105; Europe PMC 166, both from 2015 and retrieved 19 August 2026) narrow to the intersection (118 records, 99 distinct works), to the works classified as journal articles (58), to those whose full text was retrieved and coded (42), and to those that administered an instrument (27). Attrition at each step is annotated with its reason, distinguishing works that could not be retrieved from works retrieved without a vendor word form.

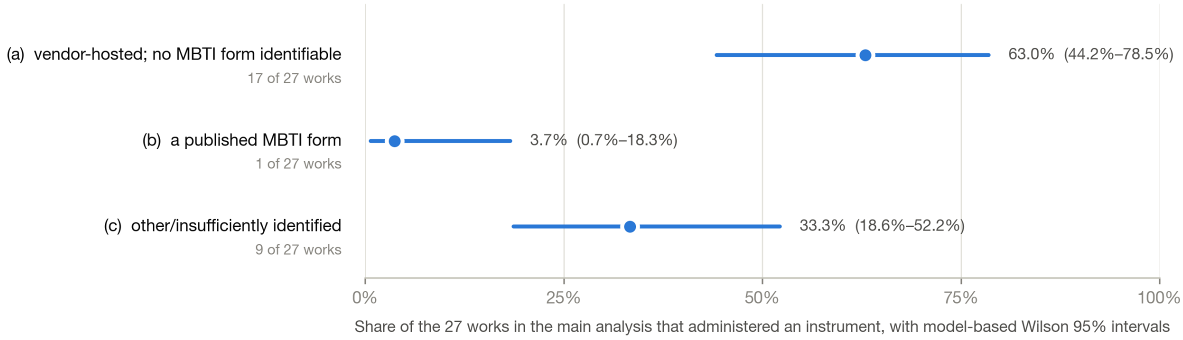


**Figure 2.** Instrument administered, among the 27 works in the main analysis that administered an instrument, with model-based Wilson 95% intervals (see Methods): (a) a vendor-hosted test with no published MBTI form identifiable, (b) a published MBTI form, (c) other or insufficiently identified, with no published MBTI form identifiable.

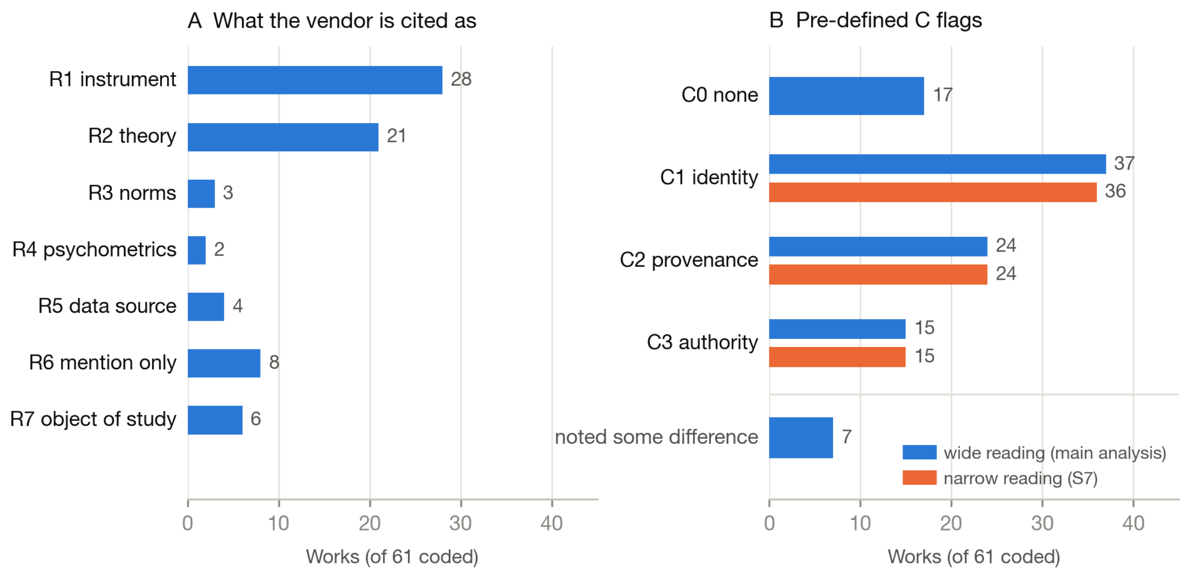


**Figure 3.** Citation-role and conflation flags across all 61 coded works, as counts. Panel A: the seven citation-role flags. Panel B: the four pre-defined C flags, with the narrow reading shown alongside the wide one, and `states_distinction` — which records that a work noted some difference between the two — displayed separately.

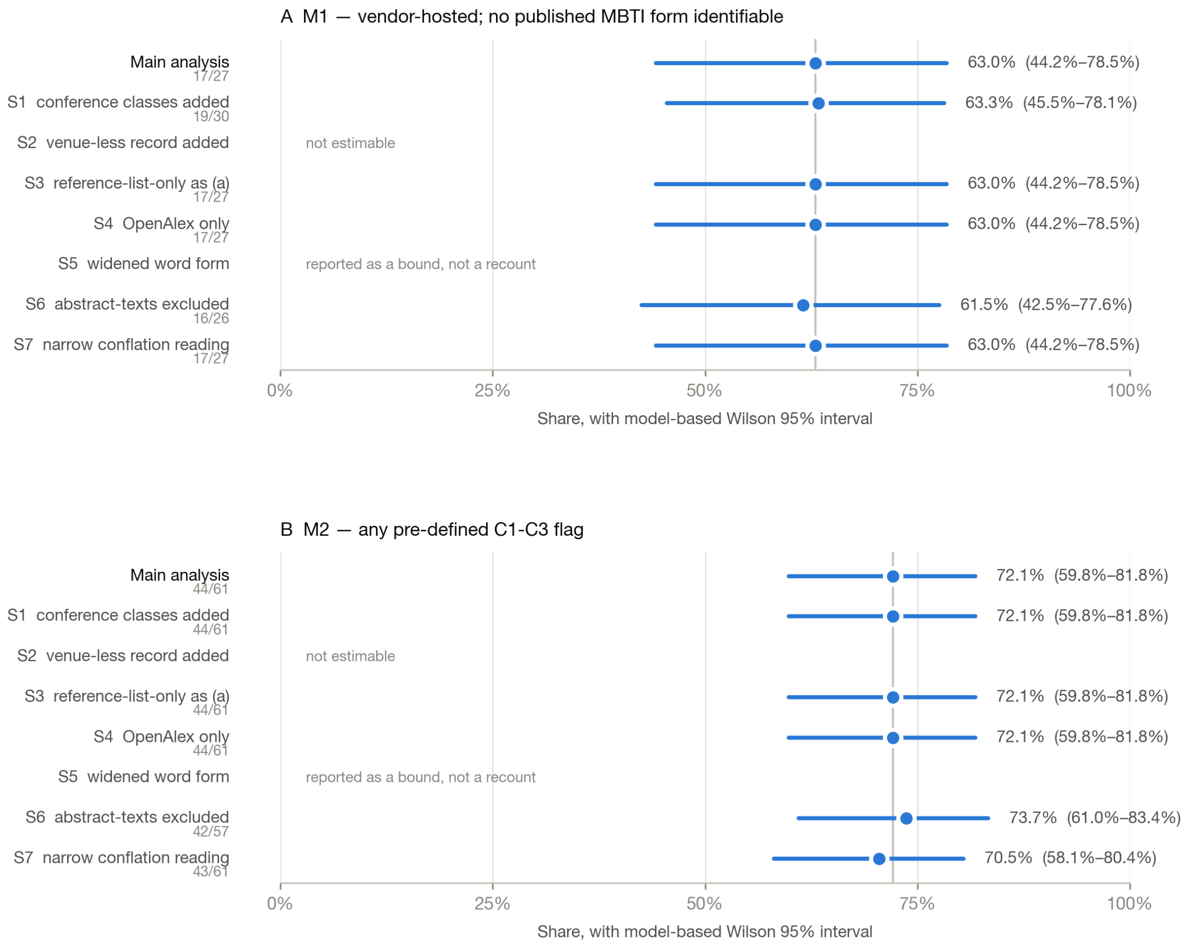


**Figure 4.** The two reported measures under the main analysis and each of the seven planned sensitivity arms, with model-based Wilson 95% intervals (see Methods). Arms that leave a measure unchanged are shown at the main-analysis value; arms that produced no estimate are annotated with the reason rather than plotted.